INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

An agent that decides, hour by hour, whether outside air can cool a data centre.

THE SITE

Location	TX facility TX_way_1530966395 -- station KDYS, America/Chicago
Committed pair	OSM 1530966391 -> 1530966394 OSM way 1530966391 / OSM way 1530966394
Facade gap	88.9 m between the two halls
Weather record	42,562 real hourly records from KDYS
Plume physics	576 steady-state solves on this site's own footprints; worst intake rise 0.3998 C at 165 deg

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day	2022-01-16 (crossing)
Plant limit	18.0 C
Notice required	3 h
Forecast skill	0.50 relative to persistence
Level anchor	sensor
Condenser bank	longest facade
Switch budget	1 mode changes per day
Minimum dwell	3 h
Max dew point	15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 6 of 24 hours with 1 mode change. The reactive incumbent took 14 hours -- 8 more -- but declared 1 unsafe hour against the agent's 0. The incumbent had to break its own switch budget 1 time to stay safe; the agent never did. 14 hours were safe but still ran chillers, because the schedule could not afford them.

Agent	6 of 24 hours free cooling, 1 mode change(s), 0 unsafe hour(s)
Incumbent	14 of 24 hours free cooling, 2 mode change(s), 1 unsafe hour(s) the incumbent broke its own switch budget 1 time(s) to stay safe 14 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	mechanical	yes	switch budget	-2.372	18.0	-3.110	1.693
01	mechanical	yes	switch budget	-2.575	18.0	-3.390	1.521
02	mechanical	yes	switch budget	-2.754	18.0	-3.720	1.429
03	mechanical	yes	switch budget	-2.933	18.0	-3.780	1.188
04	mechanical	yes	switch budget	-2.773	18.0	-3.780	1.385
05	mechanical	yes	switch budget	-3.338	18.0	-4.705	1.349
06	mechanical	yes	switch budget	-4.794	18.0	-7.705	1.297
07	mechanical	yes	switch budget	-4.312	18.0	-6.862	1.166
08	mechanical	yes	switch budget	0.772	18.0	-0.070	1.952
09	mechanical	yes	switch budget	3.912	18.0	6.137	2.512

10	mechanical	yes	switch budget	7.226	18.0	10.222	2.588
11	mechanical	yes	switch budget	11.187	18.0	13.220	1.943
12	mechanical	yes	switch budget	14.453	18.0	15.560	1.513
13	mechanical	yes	switch budget	17.624	18.0	18.330	1.458
14	mechanical	no	dry-bulb	19.227	18.0	19.440	1.423
15	mechanical	no	dry-bulb	20.505	18.0	20.280	1.407
16	mechanical	no	dry-bulb	20.914	18.0	20.220	1.115
17	mechanical	no	dry-bulb	18.895	18.0	15.553	1.620
18	FREE	yes	-	15.396	18.0	9.053	1.863
19	FREE	yes	-	12.939	18.0	6.001	1.848
20	FREE	yes	-	8.095	18.0	1.685	1.911
21	FREE	yes	-	5.942	18.0	3.220	2.061
22	FREE	yes	-	4.334	18.0	2.500	1.820
23	FREE	yes	-	1.235	18.0	0.220	1.516

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is -2.372 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

01:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is -2.575 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

02:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is -2.754 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

03:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is -2.933 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

04:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is -2.773 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

05:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is -3.338 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

06:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is -4.794 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

07:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is -4.312 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

08:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 0.772 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

09:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 3.912 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

10:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 7.226 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

11:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 11.187 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

12:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 14.452 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

13:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 17.624 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

14:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 19.227 C against a 18.0 C limit, so it fails by 1.227 C. It would take a limit 1.227 C higher, or a bound 1.227 C tighter, to change this hour.

15:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.505 C against a 18.0 C limit, so it fails by 2.505 C. It would take a limit 2.505 C higher, or a bound 2.505 C tighter, to change this hour.

16:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.914 C against a 18.0 C limit, so it fails by 2.914 C. It would take a limit 2.914 C higher, or a bound 2.914 C tighter, to change this hour.

17:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.896 C against a 18.0 C limit, so it fails by 0.896 C. It would take a limit 0.896 C higher, or a bound 0.896 C tighter, to change this hour.

18:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 15.396 C, which is 2.604 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.863 C, and the margin is measured, not chosen: 1.436 C of group-conditional forecast error for this hour of day, plus 0.4273 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

19:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 12.939 C, which is 5.061 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.848 C, and the margin is measured, not chosen: 1.549 C of group-conditional forecast error for this hour of day, plus 0.2989 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

20:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 8.095 C, which is 9.905 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.911 C, and the margin is measured, not chosen: 1.736 C of group-conditional forecast error for this hour of day, plus 0.1749 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

21:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 5.942 C, which is 12.058 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.061 C, and the margin is measured, not chosen: 1.894 C of group-conditional forecast error for this hour of day, plus 0.1670 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

22:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 4.334 C, which is 13.666 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.820 C, and the margin is measured, not chosen: 1.626 C of group-conditional forecast error for this hour of day, plus 0.1940 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

23:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 1.235 C, which is 16.765 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.517 C, and the margin is measured, not chosen: 1.337 C of group-conditional forecast error for this hour of day, plus 0.1798 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

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